

zetesis-puremath Blueprint

A Lean 4 pure-mathematics reservoir:
analytic measurability, information theory, concentration, minimax,
kernel embeddings, and dual VC.

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Chapter 1

Introduction

This blueprint pins every mathematical definition and headline theorem in `zetesis-puremath`¹ to its Lean 4 declaration. The companion source is a personal pure-math reservoir: content needed by the author’s formalization projects (`formal-learning-theory-kernel`, SMFE, and others) that is either absent from Mathlib or specialized in a way Mathlib’s current formulation does not support directly.

1.1 What is formalized here

Six substrates, each with its own chapter below:

- **Measure theory on Polish spaces.** Analytic sets are null-measurable under finite Borel measures; Kechris 30.13 capacitability theorem for analytic sets; real-valued metric total variation distance.
- **Information theory.** Real-valued Kullback-Leibler bridge to Mathlib’s `ENNReal`-valued `klDiv`; binary KL with a full derivative / monotonicity chain; Pinsker’s inequality with the sharp constant; mutual information and its zero-iff-product characterization.
- **Probability.** `BoundedRandomVariable` typeclass with a Chebyshev-via-majority bound; double-sample / `ValidSplit` / `SplitMeasure` exchangeability infrastructure; `FintypePMF` (an $\mathbb{R}$ -valued, `linarith`-compatible specialization of Mathlib’s `PMF`).
- **Decision theory.** Covering-based minimax; multiplicative-weights-update (MWU) in the Freund-Schapire $\beta = (1 - \eta)$ convention; approximate-minimax bound $T = \mathcal{O}(\log N / \varepsilon^2)$.
- **Reproducing-kernel Hilbert spaces.** Kernel mean embedding with the reproducing property; Maximum Mean Discrepancy (MMD) and characteristic kernels; Hilbert-Schmidt Independence Criterion (HSIC).
- **Combinatorics.** Assouad’s 1983 dual VC bound $\text{VCdim}(M^\top) \leq 2^{d+1} - 1$.

1.2 Verification tiers

All 114 public theorems discharge under the following tiers, all green at commit 471e5d5 (see [test/verification/CI/](#)):

- **Tier 0:** lake build — 0 errors, 0 warnings, 0 sorry in ZPM/.

¹<https://github.com/Zetetic-Dhruv/zetesis-puremath>

- **Tier 1:** `#print axioms` — 113 theorems use only `propext`, `Classical.choice`, `Quot.sound`; the 114th (`challenge_bnd_subset_cyl`) uses none.
- **Tier 2:** `leanchecker -fresh` per module (GitHub Actions `workflow_dispatch`).
- **Tier 3:** `comparator` + Landlock sandbox on GitLab CI; “*Lean default kernel accepts the solution. Your solution is okay!*” (pipeline #2463996501, duration 36 min).

Every statement in this blueprint carries a `\lean{...}` annotation linking to the Lean declaration; where the full proof is in the source, `\leanok` marks the entry as formally verified.

Chapter 2

Measure theory

2.1 Analytic measurability

Theorem 2.1 (Analytic null-measurability). *Lean: MeasureTheory.AnalyticSet.nullMeasurableSet*
 Let α be a Polish space, μ a finite Borel measure on α , and $S \subseteq \alpha$ analytic. Then S is μ -null-measurable.

Theorem 2.2 (Inner approximation by compacts). *Lean: MeasureTheory.AnalyticSet.exists_isCompact_measureReal*
 Under the hypotheses of Theorem 2.1, for every $\delta > 0$ there exists a compact $K \subseteq S$ with $\mu(S) - \mu(K) < \delta$ (in $\mathbb{R}$ -valued form).

2.2 Choquet capacity

Definition 2.3 (Choquet capacity). *Lean: MeasureTheory.IsChoquetCapacity*

A map $I : \text{Set}(\alpha) \rightarrow \mathbb{R}_{\geq 0}^{\infty}$ is a *Choquet capacity* iff (i) it is monotone, (ii) continuous from below along increasing sequences, and (iii) continuous from above along antitone sequences of closed sets. Every finite Borel measure on a Polish space is a Choquet capacity [DM78].

Definition 2.4 (Compact capacity hull). *Lean: MeasureTheory.compactCap*

For μ and S as above,

$$\text{compactCap } \mu S := \sup\{\mu(K) : K \subseteq S, K \text{ compact}\}.$$

Theorem 2.5 (Kechris 30.13: analytic capacitability). *Lean: MeasureTheory.AnalyticSet.cap_eq_iSup_isCompact*
 Let I be a Choquet capacity on a Polish space α , and S an analytic subset of α . Then

$$I(S) = \sup\{I(K) : K \subseteq S, K \text{ compact}\}.$$

The proof is the Kechris 30.13 diagonal construction on cylinders in Baire space $\mathbb{N}^{\mathbb{N}}$ [Kec95]. Together with Definition 2.4 and the fact that every finite Borel measure is a Choquet capacity, this specializes to $\mu(S) = \text{compactCap } \mu S$ (Theorem 2.1 is an immediate corollary).

2.3 Baire-space cylinder infrastructure

The proof of Theorem 2.5 requires a small theory of cylinders in $\mathbb{N}^{\mathbb{N}}$:

Definition 2.6 (Cylinder, bounded box, truncate). *Lean: Cyl, Bnd, truncate*

For $N : \mathbb{N} \rightarrow \mathbb{N}$ and $n \in \mathbb{N}$, $\text{Cyl } N n := \{g : \mathbb{N} \rightarrow \mathbb{N} \mid \forall i \leq n, g i \leq N i\}$ and $\text{Bnd } N := \bigcap_n \text{Cyl } N n$. $\text{Bnd } N = \prod_i [0, N_i]$ is compact. The map $\text{truncate } N g i := \min(g i, N i)$ sends any sequence into $\text{Bnd } N$ and agrees with g on cylinders.

The lemmas `bnd_subset_cyl`, `cyl_succ_eq`, `cyl_inter_eq_cyl_update`, `monotone_cyl_split`, `iInter_closure_image_cyl_eq`, `isCompact_bnd`, and `truncate_mem_bnd` are the technical core of the proof; they are annotated with `\lean` in the source repository and are all `\leanok`.

2.4 Total variation distance

Definition 2.7 (Real-valued total variation). `Lean: MeasureTheory.tvDistReal`

For finite Borel measures P, Q on α ,

$$\text{TV}(P, Q) := \sup_A |P(A).\text{toReal} - Q(A).\text{toReal}|$$

where the supremum is over measurable $A \subseteq \alpha$. This is the *half-sup* / metric convention (Billingsley, Villani), chosen so that Pinsker reads as $\text{TV}(P, Q) \leq \sqrt{\text{KL}(P\|Q)}/2$. The lemmas `tvDistReal_set_nonempty` and `tvDistReal_set_bddAbove` discharge the nonemptiness and boundedness conditions that $\mathbb{R}$ -valued `sSup` requires.

Remark 2.8. The repository also contains a second total-variation object, `FintypePMF.tvDistance` $p\ q = \sum_a |p\ a - q\ a|$ (the L^1 convention, see Chapter 4); the two conventions differ by a factor of 2 on discrete supports. Downstream consumers should not compose them without inserting the factor-of- $\frac{1}{2}$ correction.

Chapter 3

Information theory

3.1 Real-valued KL bridge

Definition 3.1 (Real-valued KL). *Lean: `InformationTheory.klDivReal`*

For measures P, Q on a measurable space, set

$$\text{KL}_{\mathbb{R}}(P, Q) := \begin{cases} \int \log \frac{dP}{dQ} dP & \text{if } P \ll Q, \\ 0 & \text{otherwise.} \end{cases}$$

This bridges Mathlib’s `ENNReal`-valued `klDiv` to $\mathbb{R}$ for `linarith`-compatible arithmetic; the identity $\text{KL}_{\mathbb{R}}(P, Q) = (\text{KL}(P, Q)).\text{toReal}$ holds unconditionally (`klDivReal_eq_toReal_klDiv`).

Theorem 3.2 (Non-negativity). *Lean: `InformationTheory.klDivReal_nonneg`*

$\text{KL}_{\mathbb{R}}(P, Q) \geq 0$ for all measures (Gibbs’ inequality).

Theorem 3.3 (Zero iff equal, under finiteness). *Lean: `InformationTheory.klDivReal_eq_zero_iff`*

If $P \ll Q$, $\text{KL}(P, Q) \neq \infty$, and both are finite measures, then $\text{KL}_{\mathbb{R}}(P, Q) = 0 \iff P = Q$ [[CT06](#), Thm 2.6.3].

3.2 Binary KL and Pinsker’s inequality

Definition 3.4 (Binary KL). *Lean: `InformationTheory.klBin`*

For $p, q \in [0, 1]$,

$$\text{KL}_{\text{bin}}(p, q) := p \log \frac{p}{q} + (1 - p) \log \frac{1-p}{1-q}$$

with the convention $0 \cdot \log(0/\cdot) = 0$.

Lemma 3.5 (Derivative in q). *Lean: `InformationTheory.hasDerivAt_klBin_q`*

For $q \in (0, 1)$, $\partial_q \text{KL}_{\text{bin}}(p, q) = (q - p)/(q(1 - q))$.

Theorem 3.6 (Binary Pinsker, sharp constant 2). *Lean: `InformationTheory.binary_pinsker`*

For $p \in [0, 1]$ and $q \in (0, 1)$, $2(p - q)^2 \leq \text{KL}_{\text{bin}}(p, q)$.

Theorem 3.7 (Pinsker’s inequality). *Lean: `InformationTheory.pinsker_proof`*

For probability measures $P \ll Q$ with $\text{KL}(P, Q) < \infty$,

$$\text{TV}(P, Q) \leq \sqrt{\text{KL}_{\mathbb{R}}(P, Q)/2}$$

[[CT06](#), Thm 11.6.1]. The convention matches Definition 2.7 (half-sup TV); with the L^1 convention the bound reads $\|P - Q\|_1 \leq \sqrt{2 \text{KL}}$.

Theorem 3.8 (Binary-to-general coarsening). *Lean: `InformationTheory.klBin_le_klDivReal`*

For every measurable A with $P(A) = p$, $Q(A) = q$, $\text{KL}_{\text{bin}}(p, q) \leq \text{KL}_{\mathbb{R}}(P, Q)$. This is the two-point data-processing inequality and is the step linking the binary Pinsker to the general measure-theoretic Pinsker.

3.3 Mutual information

Definition 3.9 (Real-valued mutual information). *Lean: `InformationTheory.mutualInformationReal`*

For a probability measure P on $X \times Y$ with marginals P_X, P_Y , $\text{MI}_{\mathbb{R}}(P) := \text{KL}_{\mathbb{R}}(P, P_X \otimes P_Y)$.

Theorem 3.10 (Non-negativity and independence). *Lean: `InformationTheory.mutualInformationReal_nonneg`, `InformationTheory.mutualInformationReal_eq_zero_iff_product`*

$\text{MI}_{\mathbb{R}}(P) \geq 0$; under $P \ll P_X \otimes P_Y$ and finiteness, $\text{MI}_{\mathbb{R}}(P) = 0 \iff P = P_X \otimes P_Y$.

Chapter 4

Probability

4.1 Bounded random variables + concentration

Definition 4.1 (Bounded random variable). `Lean: ProbabilityTheory.BoundedRandomVariable`

$f : \Omega \rightarrow \mathbb{R}$ is *bounded* in $[a, b]$ under μ iff f is measurable and $f(\omega) \in [a, b]$ for μ -almost every ω .

Theorem 4.2 (Chebyshev majority bound). `Lean: ProbabilityTheory.chebyshev_majority_bound`

Let $X_1, \dots, X_k$ be independent random variables with each $\mu(A_j) \geq 2/3$ where $A_j = \{X_j = \text{correct}\}$. Then for $k \geq 9/\delta$, the probability that a majority of the X_j is correct is at least $1 - \delta$. Proof via Popoviciu's variance bound + Chebyshev's inequality [BLM13, §2].

4.2 Exchangeability and double-sample

Definition 4.3 (Sample bundle). `Lean: ProbabilityTheory.ExchangeableSample`

A bundle $\langle m, \mu, \text{IsProbabilityMeasure } \mu \rangle$ where $m \in \mathbb{N}$ is the sample size and μ is a probability measure on the sample space. Used with the product measure $\mu^{\otimes 2m}$ as the natural exchangeable double-sample domain.

Definition 4.4 (Valid split). `Lean: ProbabilityTheory.ValidSplit`

$s : \text{Fin}(2m) \rightarrow \text{Bool}$ is a *valid split* iff exactly m of its values are `true`. There are $\binom{2m}{m}$ valid splits (Vapnik-Chervonenkis double-sample [VC71]).

Definition 4.5 (Uniform split measure). `Lean: ProbabilityTheory.SplitMeasure`

The uniform probability measure on the set of valid splits. This is the symmetrization measure for the double-sample trick.

4.3 Finite-type PMF

Definition 4.6 (Real-valued fintype PMF). `Lean: ProbabilityTheory.FintypePMF`

For a fintype α , a `FintypePMF` α is a pair $\langle p, h_{\geq 0}, h_{\sum=1} \rangle$ where $p : \alpha \rightarrow \mathbb{R}$, $p \geq 0$ pointwise, and $\sum_{a:\alpha} p(a) = 1$. This is an $\mathbb{R}$ -valued, `linarith`-compatible specialization of Mathlib's $\mathbb{R}_{\geq 0}^\infty$ -valued PMF [BLM13, §4].

Theorem 4.7 (Bridge to Mathlib PMF). `Lean: ProbabilityTheory.FintypePMF.toPMF_apply`

The map `toPMF` : `FintypePMF` $\alpha \rightarrow \text{PMF } \alpha$ satisfies $(\text{toPMF } p)(a) = \text{ENNReal.ofReal}(p.\text{prob } a)$, which is lossless on valid `FintypePMF` inputs.

Definition 4.8 (L^1 TV distance). *Lean: ProbabilityTheory.FintypePMF.tvDistance*

$\text{tvDistance } p \, q := \sum_{a:\alpha} |p(a) - q(a)|$. This is the *full* L^1 convention, not the half-sup TV of Definition 2.7; on discrete supports they differ by a factor of 2.

Theorem 4.9 (Basic properties). *Lean: ProbabilityTheory.FintypePMF.tvDistance_nonneg, ProbabilityTheory.FintypePMF.tvDistance_self*

tvDistance is non-negative, symmetric, and 0 on $p = q$.

Theorem 4.10 (Transfer principle). *Lean: ProbabilityTheory.FintypePMF.expectation_approx_of_tv*

For Boolean-valued f , $|\mathbb{E}_p[f] - \mathbb{E}_q[f]| \leq \text{tvDistance}(p, q)$. In the half-sup TV convention this would read $\leq 2 \cdot \text{TV}(p, q)$.

Chapter 5

Decision theory: minimax and MWU

5.1 Boolean-matrix zero-sum games

Definition 5.1 (Boolean-matrix payoff). *Lean: ProbabilityTheory.boolGamePayoff*

For a Boolean matrix $M : R \rightarrow C \rightarrow \text{Bool}$, a row distribution $p : \text{FintypePMF } R$, and a pure column $c : C$,

$$\text{boolGamePayoff } M \, p \, c := \sum_{r:R} p(r) \cdot [M \, r \, c].$$

This is the expected $\{0, 1\}$ -gain of the row player, against a pure column play, in the matrix game M [FS99].

Theorem 5.2 (One-sided minimax sanity). *Lean: ProbabilityTheory.minimax_value_le_one*

If the row player can guarantee value v against every mixed column (the *hrow hypothesis*), then $v \leq 1$.

5.2 Multiplicative weights update

Definition 5.3 (MWU config). *Lean: ProbabilityTheory.MWUConfig*

An $\text{MWUConfig } C$ is a pair $\langle w, h_{>0} \rangle$ where $w : C \rightarrow \mathbb{R}$ is the strictly positive weight vector. The *potential* is $\Phi(w) := \sum_c w(c)$, the *induced PMF* is $\text{toPMF}(w)(c) := w(c)/\Phi(w)$ [AHK12, Thm 2.1].

Definition 5.4 (Update and closed-form). *Lean: ProbabilityTheory.mwuUpdateWeights*

Given a row $r : R$ and step size $\eta < 1$, the update is $w'(c) := w(c) \cdot (1 - \eta)$ if $M \, r \, c = \text{true}$, else $w'(c) := w(c)$. This is the Freund-Schapire $\beta = (1 - \eta)$ Hedge convention [FS97]. The closed form

$$w_T(c) = w_0(c) \cdot (1 - \eta)^{\text{hitCount}_T(c)}$$

is Theorem `mwu_weight_eq_pow_hitCount`.

Theorem 5.5 (One-step potential bound). *Lean: ProbabilityTheory.potential_one_step_bound*

If the row player can guarantee value v against every mixed column and $0 \leq \eta < 1$, then after one MWU step $\Phi_{t+1} \leq \Phi_t \cdot (1 - \eta v)$.

Theorem 5.6 (T -step potential bound). *Lean: ProbabilityTheory.mwu_potential_T_bound*

Induction gives $\Phi_T \leq |C| \cdot (1 - \eta v)^T$.

5.3 Approximate minimax

Theorem 5.7 (Approximate minimax via MWU). *Lean: `ProbabilityTheory.mwu_approx_minimax`*

Fix $\varepsilon > 0$, set $\eta = \varepsilon/4$ and $T = \max\{1, \lceil 16 \log N / \varepsilon^2 \rceil\}$ where $N = |C|$. If the row player can guarantee value v against every mixed column, then the empirical row mixture produced by T rounds of MWU achieves

$$v - \varepsilon \leq \text{boolGamePayoff } M \text{ } p \text{ } c \quad \text{for every pure } c \in C.$$

This is the Freund-Schapire constructive minimax [FS99, Thm 3]; [AHK12, §3.1] unifies this with general online learning.

Theorem 5.8 (Covering-based minimax (existential)). *Lean: `ProbabilityTheory.covering_minimax`, `ProbabilityTheory.finite_approx_minimax`*

An existence counterpart to Theorem 5.7: given the same row-guarantees hypothesis, there is a mixed-row PMF with payoff at least $1/|C|$ against every pure column. This gives $v - \varepsilon \leq 1/|C|$ under a feasibility side-condition and is the logical dual of the quantitative MWU bound.

Chapter 6

Reproducing-kernel Hilbert spaces

Background conventions: the Mathlib class `RKHS` `H X` is the scalar-valued reproducing-kernel Hilbert space on a domain X , with Hilbert space $\mathcal{H} = H$ and feature map $k_x := \text{kerFun } H \ x \ 1 \in H$ satisfying $f(x) = \langle k_x, f \rangle$ [Aro50]. Throughout, $\mathcal{H}$ denotes a complete, separable, real inner product space.

6.1 Bounded kernels

Definition 6.1 (Bounded kernel). *Lean: `BoundedKernel`*

A `BoundedKernel` $\mathcal{H}$ is a witness $\langle C, h_{\geq 0}, h_{\leq} \rangle$ with $C \geq 0$ and $\|k_x\| \leq C$ for all $x : X$. Bounded-feature kernels are the canonical setting for kernel mean embeddings [SC08, §4.2].

Theorem 6.2 (Evaluation bound). *Lean: `norm_apply_le_bound_mul_norm`*

$|f(x)| \leq C \cdot \|f\|$ for every $f \in \mathcal{H}$ and $x \in X$.

6.2 Kernel mean embedding

Definition 6.3 (Kernel mean embedding). *Lean: `kernelMeanEmbedding`*

For a measure P on X , set

$$\mu_P := \int_X k_x \, dP(x) \in \mathcal{H}$$

(Bochner integral) [SGSS07].

Theorem 6.4 (Reproducing property for the embedding). *Lean: `kernelMeanEmbedding_reproducing`*

For integrable f , $\langle \mu_P, f \rangle = \int_X f(x) \, dP(x)$ [GBR⁺12, §2, eq. (2)].

Theorem 6.5 (Norm bound). *Lean: `kernelMeanEmbedding_norm_le`*

Under *IsProbabilityMeasure* P , $\|\mu_P\| \leq C$.

6.3 Maximum Mean Discrepancy

Definition 6.6 (Squared MMD). *Lean: `mmdSq`*

$\text{MMD}^2(P, Q) := \|\mu_P - \mu_Q\|^2$ [GBR⁺12, Defn. 2].

Theorem 6.7 (Metric axioms mod characteristic). *Lean: `mmdSq_nonneg`, `mmdSq_self`, `mmdSq_comm`*

$\text{MMD}^2 \geq 0$, $\text{MMD}^2(P, P) = 0$, and MMD^2 is symmetric in its arguments.

Definition 6.8 (Characteristic RKHS). *Lean: IsCharacteristic*

$\mathcal{H}$ is *characteristic on X* iff the kernel mean embedding $P \mapsto \mu_P$ is injective. This is the modern definition [SGF⁺10, Defn. 6].

Theorem 6.9 (Zero-iff-equal under characteristic kernel). *Lean: mmdSq_zero_iff*

Under *IsCharacteristic* $\mathcal{H}$, $\text{MMD}^2(P, Q) = 0 \iff P = Q$.

6.4 Hilbert-Schmidt Independence Criterion

Definition 6.10 (HSIC via product-space MMD). *Lean: hsicDef*

For a joint measure P on $X \times Y$ with marginals P_X, P_Y ,

$$\text{HSIC}(P) := \text{MMD}^2(P, P_X \otimes P_Y).$$

This is the modern unified formulation [SGF⁺10, Thm 24]; originally introduced as a cross-covariance-operator Hilbert-Schmidt norm [GBSS05].

Theorem 6.11 (Zero-iff-independent). *Lean: hsicDef_nonneg, hsicDef_eq_zero_of_independent, hsicDef_zero_iff_i*

$\text{HSIC}(P) \geq 0$; $P = P_X \otimes P_Y$ implies $\text{HSIC}(P) = 0$; conversely, under characteristic kernels on $X \times Y$, $\text{HSIC}(P) = 0 \iff P = P_X \otimes P_Y$.

Chapter 7

Combinatorics: dual VC

7.1 Binary matrix setup

Definition 7.1 (Binary matrix). *Lean: BinaryMatrix*

$\text{BinaryMatrix } m\ n := \text{Fin } m \rightarrow \text{Fin } n \rightarrow \text{Bool}$. Type-definitionally equal to $\text{Matrix } (\text{Fin } m) (\text{Fin } n) \text{ Bool}$.

Definition 7.2 (Shattering). *Lean: BinaryMatrix.shatters, BinaryMatrix.toFinsetFamily*

For a binary matrix M and $S \subseteq \text{Fin } n$, M *shatters* S iff every pattern $t \subseteq S$ is realized by some row. The induced set family $\text{toFinsetFamily } M$ consists of the true-supports of the rows of M [VC71, Sau72].

Theorem 7.3 (Consistency with Finset.Shatters). *Lean: BinaryMatrix.shatters_iff*

M *shatters* $S \iff \text{toFinsetFamily } M$ *shatters* S in Mathlib’s sense.

7.2 Sauer-Shelah corollary

Theorem 7.4 (Cardinality of the induced family). *Lean: BinaryMatrix.card_toFinsetFamily_le*

If $\text{VCdim}(\text{toFinsetFamily } M) \leq d$ then $|\text{toFinsetFamily } M| \leq \sum_{k=0}^d \binom{n}{k}$. This is the Sauer-Shelah bound [Sau72, VC71] routed through Mathlib’s $\text{Finset.card_shatterer_le_sum_vcDim}$ (Pajor + Sauer-Shelah) into the matrix setting.

7.3 Assouad’s dual VC bound

Lemma 7.5 (Dual-to-primal shattering coding). *Lean: BinaryMatrix.transpose_shatters_imp_shatters*

If $M^\top$ *shatters* $S \subseteq \text{Fin } m$ with $|S| \geq 2^{d+1}$, then M *shatters* some $T \subseteq \text{Fin } n$ with $|T| = d + 1$. Proof by a bitstring-coding argument: over $(d+1)$ -bit patterns on S , each pattern must be realized by a distinct column, giving $d + 1$ distinguishing columns that M *shatters* [Ass83, Thm 2.13].

Theorem 7.6 (Assouad’s bound: $\text{VCdim}(M^\top) \leq 2^{d+1} - 1$). *Lean: BinaryMatrix.assouad_transpose_vcDim*

If $\text{VCdim}(M) \leq d$ then $\text{VCdim}(M^\top) \leq 2^{d+1} - 1$ [Ass83, Thm 2.13]. The bound is tight at small d ; no strengthening is known for general d without additional structure.

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